

# Known Issues and Interpretation Notes

*Pipeline handover documentation — occlusion-resilient tracking implementation*

This document lists everything found during implementation and testing that is relevant to interpreting the results correctly. Nothing here is hidden or minimized — the goal is that the results can be trusted precisely because the limitations are stated plainly.

## Bugs Found and Fixed During Implementation

### 1. Depth pipeline library API mismatch

The HuggingFace transformers depth-estimation pipeline's 'depth' output key returns an already-normalized 0-255 image in the installed library version, not the raw float depth values the design's locked thresholds (P25=1.06, P65=3.61) assume. The raw values are under a different key, 'predicted\_depth'. This did not crash — it silently produced incorrect depth bins (everything classified 'far') until caught by direct inspection of the pipeline's output.

Fixed in src/depth.py.

### 2. LLM enrichment timeout too short

The configured timeout (5.0s) did not leave enough margin for the actual reasoning prompt, which reliably takes 3.4-3.7s. This caused repeated timeout failures under normal conditions. The designed fallback behavior worked correctly throughout — the rule-based reasoning output was never lost, only the optional LLM text.

Fixed in config.yaml (llm\_timeout\_seconds: 5.0 -> 15.0).

### 3. Track history could not capture pre-loss stability duration

A design-time gap: the per-track state correctly resets its stability counter the instant a track is lost, but this meant the BS-1 trigger had no way to check how long a track had been stable before disappearing. Caught during design, before it caused any incorrect behavior.

Fixed by capturing this value at the exact loss transition, before the counter resets.

## Known Limitations

### 4. DeepSORT does not always re-identify a person after full occlusion

Testing on a known occlusion event (a person passing behind a clothing rack, a 26-frame / ~1.7s gap) showed the person receiving a new track ID on reappearance rather than recovering the original one, even though the configured `max_age` (30 frames) exceeds the gap length. This suggests an appearance-matching limitation on reappearance, not simply a timeout.

The BS-1 reasoning trigger was deliberately designed to fire on the loss of a track, not on whether it is later successfully re-identified — this limitation does not affect BS-1's correctness.

### 5. google-generativeai is a deprecated library

Google has ended support for this package in favor of a newer one, `google-genai`. The current implementation works correctly (validated against the live API), but this is technical debt for future maintenance of this codebase.

### 6. Gemini free-tier daily quota is easily exhausted during testing

The free API tier allows 20 requests/day. Repeated testing during implementation exhausted this quota multiple times, each time triggering the designed fallback (rule-based reasoning output alone, no LLM text, no crash). This is expected behavior under quota limits, not a defect.

A paid tier or a fresh daily quota is recommended for a demonstration requiring uninterrupted LLM enrichment.

### 7. Track fragmentation may affect the raw BS-1 event count on any given clip

In a crowded scene, tracker output can fragment one real person's continuous presence into multiple short-lived track IDs. Since each qualifying track independently triggers its own BS-1 evaluation, it is possible for a busy clip's total event count to include more than one event per real occlusion situation, if the same person's track fragmented beforehand.

The trigger logic itself has been verified correct through dedicated testing; this note is about interpreting a raw event count from any one clip as an exact count of distinct real-world situations, which has not been individually cross-checked frame-by-frame.

## Interpretation Notes

### 8. The two demo clips demonstrate two different things

The Test6 clip comes from the same recording sessions used to build the training dataset. It demonstrates that the BS-1 trigger fires correctly on a known occlusion pattern — a functional correctness check, not a generalization claim.

The second demo clip was extracted from a recording never used in dataset construction, training, or any prior evaluation. It demonstrates that the full pipeline generalizes to genuinely unseen, real-world footage — this is the generalization evidence.

### 9. This handover's results are separate from the thesis's Chapter 6 evaluation

The thesis's validated detection and tracking metrics come from a specifically-constructed, leakage-free held-out evaluation set. Neither demo clip in this handover is that evaluation set — they are separate evidence, produced for this pipeline demonstration.

### 10. Implementation status relative to the submitted thesis

The submitted thesis states that the depth annotation, track-depth association, and BS-1 reasoning components were specified but not implemented or evaluated at the time of submission. That was accurate at the September 8th submission date.

Since then, all three components have been implemented, tested, and validated, as reflected in this handover and the accompanying codebase.

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Full technical detail for each item is available in the codebase's docstrings and commit history. This document summarizes what is relevant for interpreting the delivered pipeline's results correctly.